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# Automated detection and numbering of primary and permanent teeth in digital impressions of children using artificial intelligence

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ABSTRACT

*Objectives:* Considering the importance of distinguishing between primary and permanent teeth in children with mixed dentition, this study aimed to develop and evaluate an automated method for segmenting and labelling primary and permanent teeth in digital impressions.
*Methods:* 716 digital impressions from 351 patients with primary or mixed dentitions were collected from the Netherlands, Brazil, and the 3DTeethSeg22 challenge dataset. The scans were annotated with tooth segmentations and primary and permanent teeth FDI numbers. A deep learning model was applied that combined large-context predictions for tooth labelling with high-resolution predictions for tooth segmentation. Using the collected scans, the model was trained and evaluated with five-fold cross-validation for tooth detection (F1-score), tooth segmentation (Dice score), and tooth labelling (macro-F1). Additionally, the model was trained and evaluated using the train-test split of the 3DTeethSeg22 challenge dataset.
*Results:* The developed model achieved highly effective results for tooth detection (F1-score = 0.996), tooth segmentation (Dice = 0.969), and tooth labelling (macro-F1 = 0.989). Moreover, a digital impression was processed in under two seconds on average. Furthermore, the proposed method outperformed the top-ranked 3DTeethSeg22 challenge submission (score = 0.954 vs. 0.976) and was particularly effective for tooth labelling (tooth identification rate = 0.910 vs. 0.955). Failure cases revealed mistakes for unusual dental conditions or ambiguous tooth eruption patterns.
*Conclusions:* A highly effective algorithm for tooth segmentation was developed to differentiate between primary and permanent teeth in digital impressions. This fast and accurate model can benefit dentists in documenting children's teeth during the mixed dentition stage.
*Clinical significance:* The algorithm provides an accurate and reliable tool for AI-assisted identification and numbering of primary and permanent teeth in digital impressions obtained from children with mixed dentition, thereby enhancing clinical workflow, improving treatment planning accuracy, and facilitating communication with patients and caregivers.

## 1. Introduction

Digital dentistry is revolutionising clinical practice, offering significant advancements in diagnosis, treatment planning, and patient care [1]. This transformation is also becoming a reality in paediatric dentistry. Intraoral scanners (IOS), in particular, are increasingly used as an accurate alternative to traditional impression methods, improving patient comfort and enhancing digital workflows. IOS provide a reliable

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solution for capturing precise digital impressions [2], which have been utilised for diagnosing caries lesions [3], monitoring erosive tooth wear [4], supporting orthodontic treatment decisions [5], and designing indirect restorations [6].

More recently, digital dentistry has been further enhanced by Artificial Intelligence (AI), resulting in more accurate and efficient workflows that accelerate several clinical steps [7]. AI, particularly deep learning, can process large amounts of annotated data and learns to replicate expert-level performance [8]. This technology has been applied to digital impressions for automated caries detection [9], tooth wear monitoring [10], detection and classification of dental restorations [11], and aligning digital impressions in maximum intercuspal position [12]. Many digital workflows supported by AI require effective methods for tooth segmentation and labelling as a foundational step before further processing [13,14].

However, in paediatric dentistry, children with mixed dentition present additional challenges for clinicians. This transitional stage is characterised by the simultaneous presence of primary and permanent teeth, resulting in greater anatomical variability and increased occlusal complexity [15]. Accurate identification of tooth type, position, and eruption chronology is crucial for a precise diagnosis of dental caries, malocclusions, and dental anomalies (such as supernumerary teeth or anodontia), providing a more effective basis for treatment planning, including restorative, orthodontic, and surgical management [16–19]. Therefore, AI-powered tools designed for clinical applications in children should be capable of distinguishing between primary and permanent teeth when used in cases of mixed dentition. Previous studies have presented accurate deep learning solutions for identifying different types of teeth in panoramic radiographs [16,17,19] and Cone Beam Computed Tomography (CBCT) [18] of children with mixed dentition. Nevertheless, to the best of our knowledge, no method or publicly available dataset has been released for this purpose using digital impressions obtained from IOS.

Previously developed algorithms often relied on the relative positions of teeth to assign unique labels according to the FDI tooth numbering system. As a primary tooth and its corresponding permanent tooth occupy the same position in a dental arch, an effective algorithm to recognise a tooth as primary or permanent must consider the tooth's size, shape, wear characteristics, and the eruption status of surrounding teeth. Therefore, the present study aimed to develop and evaluate a deep learning-based method for automated tooth instance segmentation and labelling in digital impressions obtained from IOS of children, with the capacity to differentiate between primary and permanent teeth.

## 2. Material and methods

This work followed the ethical principles of the World Medical Association (Declaration of Helsinki), and ethical approval for using patient data from the Netherlands and Brazil has been granted (METC Oost-Nederland, file number 2024–17456; Research Ethics Committee of the School of Dentistry at the University of São Paulo, file number 70690623.1.0000.0075). The checklist for AI research in dentistry has been consulted for the reporting of this study [20].

### 2.1. Data

The current study includes a convenience sample of 351 patients and 716 digital impressions with a primary or mixed dentition, collected from the Department of Dentistry, Radboud University Medical Center, Nijmegen, the Netherlands (106 patients, 226 scans), the Department of Pediatric Dentistry, School of Dentistry, University of São Paulo, São Paulo, Brazil (48 patients, 96 scans), and the publicly available 3DTeethSeg22 challenge dataset (197 patients, 394 scans) [20]. This collection was acquired using different IOS, and patient ages range from 6 to 16 years, with no discrepancy in gender attributed at birth (Table 1). Exclusion criteria involved: (i) subjects without primary teeth; (ii) with third molars; (iii) patients with extracted premolars; (iv) and patients where the digital impression of one dental arch was missing. The digital impressions made in participants from Nijmegen were acquired *in vitro* by scanning plaster casts collected for a longitudinal research project. For the other datasets, digital impressions were acquired *in vivo*. The scans were converted to PLY format without colour information, as scans acquired from Nijmegen and the 3DTeethSeg22 challenge dataset were missing colour. All data were pseudonymised before analysis.

**Table 1**
Dataset characteristics of the digital impressions from Nijmegen, São Paulo, and the public 3DTeethSeg22 challenge dataset.

| | Nijmegen | São Paulo | 3DTeethSeg22 |
|---|---|---|---|
| **Patients** | 106 | 48 | 197 |
| **Scans** | 226 | 96 | 394 |
| **Intraoral scanner(s)** | 3Shape TRIOS 3 | Dentsply Sirona Omnicam | Dentsply Sirona Primescan; 3Shape TRIOS 3; iTero Element™ 2 Plus |
| **Age** | 6 to 8 years old | 8 to 12 years old | Under 16 years old* |
| **Gender attributed at birth** | 54 male, 52 female | 24 male, 24 female | 50 % male, 50 % female* |

*Reported by Ben-Hamadou et al. (2022) [22].

### 2.2. Data annotation

All teeth in the collected digital impressions were annotated with a point-wise segmentation and a label following the FDI numbering system with the 3DMedX software (3DLab, Nijmegen, the Netherlands) based on a 3-step process. First, pre-annotations with only FDI numbers of permanent teeth were collected from the 3DTeethSeg22 challenge dataset [21] or produced by the StratifiedTSegNet algorithm [22]. Then, a dentist reviewed and revised the pre-annotations, introducing FDI numbers of primary teeth. Lastly, a PhD student and two dentists reviewed and revised the annotations in joint sessions to reach a consensus. The scans collected from Nijmegen predominantly consisted of primary dentitions, whereas those collected from São Paulo and the 3DTeethSeg22 challenge dataset were from children with mixed dentition (Table 2). The order in which the teeth erupted corresponded with other reports of tooth eruption on most occasions [23]. All the FDI numbers are well represented, with the lowest prevalence of 18 % for the permanent upper second molars.

### 2.3. Deep learning model

To accurately assign a tooth label, distinguishing between primary and permanent teeth involves evaluating factors such as tooth size, shape, and the surrounding dentition. This requires an automated method that considers a broad context and integrates information from the complete digital impression to ensure accurate labelling.

Conversely, precise tooth segmentation primarily depends on the characteristics of the tooth itself. In this case, a high resolution is more critical than a broad context, as fine details directly influence segmentation accuracy.

These insights have guided the development of ToothInstanceNet, a two-stage deep-learning model designed for tooth instance segmentation and labelling, with the aim of predicting tooth landmarks [24]. The model effectively combines large-context predictions for labelling and high-resolution predictions for segmentation and tooth landmarks, optimising performance for all tasks. The current work introduces several optimisations to ToothInstanceNet to boost its effectiveness for tooth labelling. In particular, for mixed dentitions, digital impressions with uncommon combinations of FDI numbers are oversampled during training, and the number of labels per quadrant is increased by five to include the five possible primary teeth. Fig. 1 shows the overview of the

**Table 2**
Counts and prevalences of the annotated tooth labels based on the digital impressions from Nijmegen, São Paulo, and the public 3DTeethSeg22 challenge dataset. A total of 8321 teeth were annotated.

| | | Nijmegen Count (%) | São Paulo Count (%) | 3DTeethSeg22 Count (%) | Total Count (%) |
|---|---|---|---|---|---|
| **Permanent upper teeth** | Label 11,21 | 1 (0.44) | 73 (76) | 394 (100) | 468 (65) |
| | Label 12,22 | 0 (0) | 54 (56) | 384 (97) | 438 (61) |
| | Label 13,23 | 0 (0) | 7 (7.3) | 255 (65) | 262 (37) |
| | Label 14,24 | 0 (0) | 28 (29) | 317 (80) | 345 (48) |
| | Label 15,25 | 0 (0) | 12 (13) | 149 (38) | 161 (22) |
| | Label 16,26 | 32 (14) | 75 (78) | 393 (100) | 500 (70) |
| | Label 17,27 | 0 (0) | 0 (0) | 131 (33) | 131 (18) |
| **Permanent lower teeth** | Label 31,41 | 5 (2.2) | 79 (82) | 394 (100) | 478 (67) |
| | Label 32,42 | 0 (0) | 73 (76) | 394 (100) | 467 (65) |
| | Label 33,43 | 0 (0) | 26 (27) | 327 (83) | 353 (49) |
| | Label 34,44 | 0 (0) | 21 (22) | 317 (80) | 338 (47) |
| | Label 35,45 | 0 (0) | 12 (13) | 140 (36) | 152 (21) |
| | Label 36,46 | 23 (10) | 77 (80) | 393 (100) | 493 (69) |
| | Label 37,47 | 0 (0) | 1 (1.0) | 158 (40) | 159 (22) |
| **Primary upper teeth** | Label 51,61 | 223 (99) | 21 (22) | 0 (0) | 244 (34) |
| | Label 52,62 | 226 (100) | 33 (34) | 0 (0) | 259 (36) |
| | Label 53,63 | 226 (100) | 83 (86) | 103 (26) | 412 (58) |
| | Label 54,64 | 224 (99) | 66 (69) | 74 (19) | 364 (51) |
| | Label 55,65 | 224 (99) | 82 (85) | 239 (61) | 545 (76) |
| **Primary lower teeth** | Label 71,81 | 222 (98) | 15 (16) | 1 (0.25) | 238 (33) |
| | Label 72,82 | 225 (100) | 19 (20) | 2 (0.51) | 246 (34) |
| | Label 73,83 | 226 (100) | 66 (69) | 52 (13) | 344 (48) |
| | Label 74,84 | 225 (100) | 69 (72) | 77 (20) | 371 (52) |
| | Label 75,85 | 222 (98) | 80 (83) | 254 (64) | 556 (78) |
| **Total** | | 2304 | 1072 | 4945 | 8321 |

method and Supplementary Section 1 provides additional details.

Stage 1 implements tooth instance segmentation and labelling using an end-to-end deep learning model inspired by Wang et al. (2023) [25]. A digital impression is pre-processed to a low-resolution point cloud, so the complete scan can be processed. This point cloud is processed by a model with one shared encoder and three separate decoders.

The first decoder results in a seed map, identifying the points belonging to teeth and predicting a higher seed value for points closer to a tooth centre. The second decoder predicts the x-, y-, and z-coordinate offsets from each point to the closest tooth centre and the three-dimensional size or bandwidth of that tooth. The predictions from the first two decoders are used to identify tooth instances in an iterative fashion as follows. The point with the highest seed value is selected, and a Gaussian distribution is defined with the point's proposed tooth centre and bandwidth as mean and diagonal covariance matrix, respectively. Subsequently, the proposed tooth centre of all points are used to compute the probability of each point belonging to the new tooth instance, given the Gaussian distribution. Finally, points above a threshold probability are selected as the new tooth instance. This process is repeated for points not yet selected until no high seed value remains, resulting in a tooth instance segmentation with each point attributed to a unique tooth instance or the background.

The third decoder learns informative features for tooth labelling and computes a tooth instance embedding by averaging over the features of the instance's points (masked average pooling). This embedding is processed by a multi-layer perceptron (MLP) to predict one of twelve tooth labels; the first seven permanent teeth and the five primary teeth in each quadrant. The tooth labels are converted to FDI numbers using the dental arch (mandible, maxilla), and the left and right teeth are differentiated based on the position of the incisors, resulting in a tooth instance segmentation with FDI numbers.

Stage 2 pre-processes the digital impression to a high-resolution point cloud and determines a tooth crop by selecting the 10,000 points closest to a tooth centre. This tooth crop is processed by an encoder followed by a decoder to predict a binary segmentation of the centre tooth. Stage 2 determines a high-resolution segmentation of each tooth instance and interpolates the predictions to the vertices of the digital impression. Finally, the predictions are aggregated, and a vertex segmented as a tooth is attributed an FDI number based on the tooth instance with the highest segmentation confidence.

### *2.4. Model inference*

The pre-processed point cloud and a horizontally flipped version were used to determine two tooth instance segmentations using stage 1 (test-time augmentation). Non-maximum suppression (NMS) was applied to remove redundant tooth instances, and high-resolution tooth crops were determined around each identified tooth centre. Based on these crops, stage 2 was run to determine binary segmentations of the centre teeth and NMS was applied again. The tooth centres were updated based on the binary segmentations to determine tooth crops where the identified tooth instances were more centred. These crops were processed by stage 2 and NMS again and aggregated to determine an instance number and a foreground probability for each vertex of the digital impression. A graph-cut algorithm was applied to the foreground probabilities and the edges of the digital impression to determine the foreground tooth points [26]. Finally, a tooth instance segmentation was determined by masking the instance numbers using the foreground points and the FDI numbers were predicted using the features from the third decoder of stage 1 masked by the final tooth instance segmentation.

### *2.5. Model evaluation*

The patients included in the current study were divided over five folds using multi-label stratification based on the FDI numbers and the method was trained using five-fold cross-validation [27], such that all digital impressions could be used for training and evaluation. The predictions and annotations were compared for tooth detection, tooth segmentation, and tooth labelling. Furthermore, a qualitative evaluation of the results is presented with examples of digital impressions, annotations, and predictions, as well as several failure cases. Moreover, an ablation study was performed to investigate the utility of running stage 1 with test-time augmentation, adding stage 2, running stage 2 twice, and the graph-cut algorithm.

Lastly, the model was trained on the train split of the 3DTeethSeg22 challenge [22] and evaluated on the original test split, to compare the effectiveness of our method to other challenge submissions. This model used the first seven permanent teeth of a quadrant as tooth labels, and identified the third molars by a pair of second molars in one quadrant. During the annotation process of the current study, some cases in the train split were revised to correct missing teeth, tooth segmentations

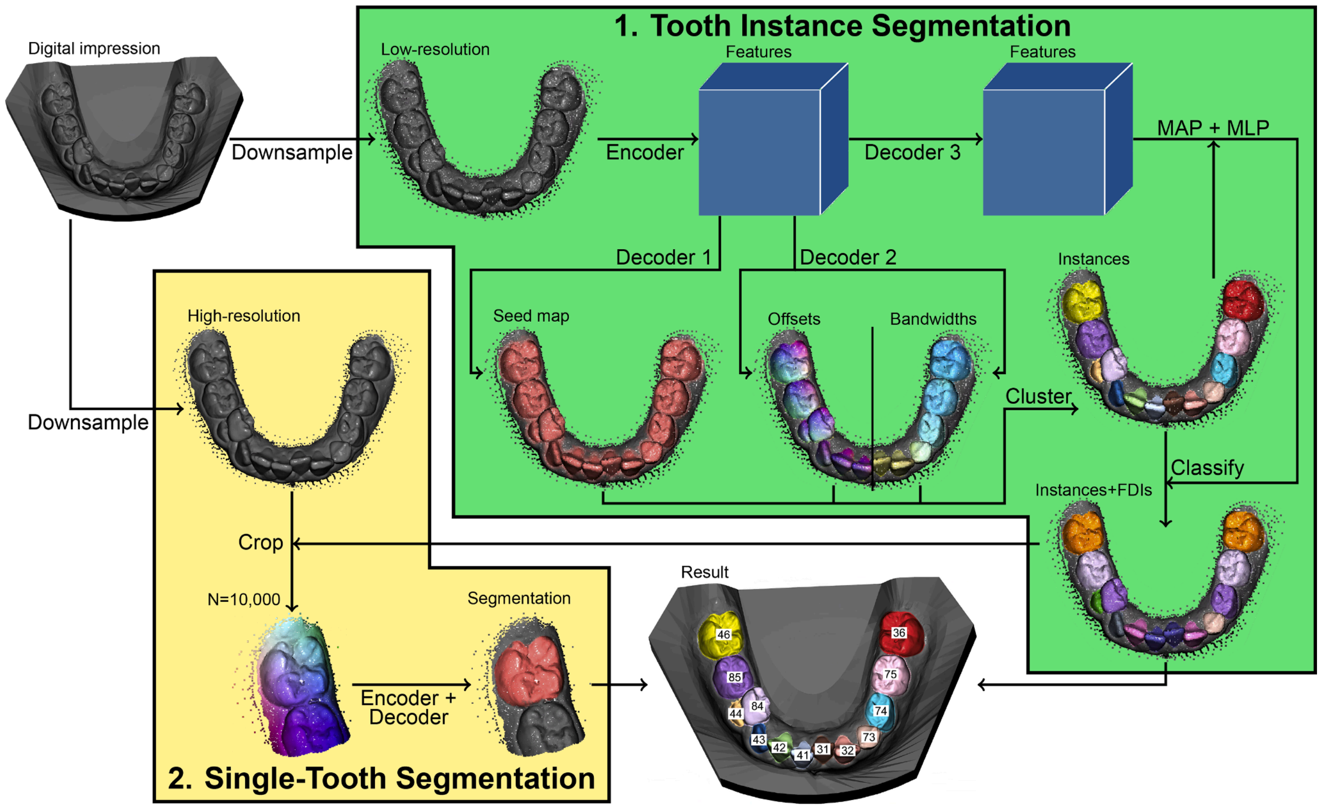


**Fig. 1.** Deep learning model with a tooth instance segmentation stage and a single-tooth segmentation stage [24]. A digital impression is first downsampled to a low-resolution point cloud to predict a seed map, offsets, and bandwidths of the complete scan. These are used to produce a tooth instance segmentation, where the features of each instance are processed further to label it with an FDI number. Then, single-tooth crops are taken from a high-resolution point cloud to predict a binary segmentation of each detected tooth, which are combined with the FDI numbers for the final result. MAP = masked average pooling. MLP = multi-layer perceptron.

that included gingiva, or incorrect FDI numbers.

The challenge submissions included Chompers, which implemented a two-stage model for tooth centre prediction and single-tooth segmentation and labelling based on point clouds. OS predicted tooth centres and labels from a 2D top-view of the digital impression and cropped the 3D mesh to predict single-tooth segmentations, followed by a graph-cut algorithm. TeethSeg predicted a coarse multi-class segmentation and revised the tooth-tooth and tooth-gingiva boundaries using a random walker with local geometric features. IGIP first predicted tooth centres and a binary segmentation of all teeth to extract crops with shape, curvature, and position features. These were used to predict single-tooth segmentations and labels, after which label errors were corrected during post-processing. FiboSeg rendered multiple 2D views of a digital impression and predicted multi-class segmentations, which were projected back to the scan with majority voting, followed by removal of redundant teeth and smoothing of tooth-gingiva boundaries. Finally, CGIP sampled a point cloud from the digital impression and predicted tooth centres and a multi-class segmentation, which were combined into a tooth instance segmentation with labels. This result was refined by predicting tooth-gingiva boundaries in single-tooth crops, and by performing the predictions again for a point cloud with more points at the tooth boundaries.

### 2.6. Statistical analysis

Statistical analyses were performed using scikit-learn (v. 1.5.1 [28]). Tooth detection was evaluated by matching predicted and annotated teeth with a point-wise intersection over union (IoU) $= \frac{TP}{TP+FP+FN}$ of at least 0.5 and computing tooth-wise F1-score (*i.e.* Dice score) $= \frac{2*TP}{2*TP+FP+FN}$, where $TP$, $FP$, and $FN$ are true-positive, false-positive, and false-negative detections, respectively. Tooth segmentation was investigated by the point-wise Dice score between matched teeth and tooth labelling was evaluated with macro-F1 for matched teeth, as well as confusion matrices. The overall effectiveness was evaluated using macro-IoU, which averages the point-wise IoU values of each FDI number.

The 3DTeethSeg22 challenge evaluated tooth detection with the teeth localisation accuracy (TLA), defined as the mean of tooth-size-normalised Euclidean distances between annotated teeth and the closest predicted tooth. Furthermore, it defines teeth segmentation accuracy (TSA) as the F1-score comparing all annotated and predicted tooth points. Lastly, tooth labelling is evaluated using the tooth identification rate (TIR), defined as the percentage of annotated teeth with a close predicted tooth that is labelled with the same FDI number. Overall effectiveness was evaluated by averaging these metrics.

## 3. Results

The developed method demonstrated high accuracy for tooth detection, achieving an F1-score exceeding 0.995 for cases from the 3DTeethSeg22 challenge dataset and those involving a primary or mixed dentition (Table 3). Similarly, it produced highly effective results for tooth segmentation and labelling. Notably, ToothInstanceNet excelled in tooth segmentation for the 3DTeethSeg22 challenge dataset and in tooth labelling for mixed digital impressions. The algorithm processed a digital impression in under two seconds, showcasing its computational efficiency. The ablation study highlighted that stage 2 significantly enhanced performance across all tasks, while post-processing provided an improvement, resulting in the most effective overall outcomes (Table 3).

**Table 3**
Results of ablation study for the deep learning model trained and evaluated on the complete 3DTeethSeg22 challenge dataset or the digital impressions with primary and permanent teeth. Best metrics are highlighted bold.

| Dataset | Method | F1-score | Tooth Dice | Tooth macro-F1 | macro-IoU | Time (s) |
|---|---|---|---|---|---|---|
| **3DTeethSeg22** | Stage 1 | 0.9909 | 0.9608 | 0.9528 | 0.8334 | 0.352 |
| | + test-time aug | 0.9856 | 0.9606 | 0.9542 | 0.8350 | 0.457 |
| | + stage 2 | 0.9955 | 0.9818 | 0.9590 | 0.8744 | 1.008 |
| | + stage 2 twice | 0.9963 | 0.9818 | **0.9598** | 0.8780 | 1.401 |
| | + post-processing | **0.9964** | **0.9824** | **0.9598** | **0.8790** | 1.707 |
| **Mixed dentitions** | Stage 1 | 0.9898 | 0.9500 | **0.9894** | 0.8779 | 0.325 |
| | + test-time aug | 0.9791 | 0.9501 | 0.9680 | 0.8773 | 0.437 |
| | + stage 2 | 0.9960 | 0.9687 | 0.9884 | 0.9083 | 0.861 |
| | + stage 2 twice or | 0.9960 | 0.9688 | 0.9884 | 0.9058 | 1.217 |
| | post-processing | **0.9961** | **0.9692** | 0.9885 | **0.9092** | 1.421 |

Average inference time per scan is reported.

ToothInstanceNet was highly effective for the 3DTeethSeg22 challenge, outperforming the most effective challenge submission by a substantial margin (score = 0.9539 vs. 0.9764, Table 4), setting new benchmark results for tooth instance segmentation and labelling. Equivalently, the deep learning model demonstrated very high effectiveness for cases involving primary teeth, even surpassing the efficacy for the 3DTeethSeg22 challenge, primarily due to highly effective tooth labelling (Table 3). The confusion matrices in Fig. 2 further highlight the excellent effectiveness of tooth labelling in mixed digital impressions, as only a few mistakes off the diagonal are visible. In particular, the final method was equally effective for maxillary and mandibular teeth, although it occasionally confused primary and permanent canines.

Qualitative results (Fig. 3) underscore the robustness of the proposed method in handling complex dental conditions and various stages of tooth eruption. For instance, in the third digital impression, an incorrect tooth label from stage 1 (FDI 34) was corrected in stage 2 to FDI 35, as masked average pooling (MAP) leveraged a more accurate tooth segmentation. Additionally, post-processing improved the tooth-gingiva boundaries in the fourth digital impression, ensuring precision even beyond the reference annotations. These results highlight the significant improvements in segmentation accuracy achieved through the integration of additional model components.

Three examples of failure cases are illustrated in Fig. 4, including a false-positive detection (Fig. 4A), a false-negative detection (Fig. 4B), and incorrect tooth labels (Fig. 4C). In the first case, the digital impression lacked teeth on one side, which puzzled the algorithm. The second case involved an unusual tooth eruption sequence, and the algorithm did not detect the second permanent molar (FDI 47) in its very early eruption stage. Fig. 4C presents a highly atypical dental condition with several cases of prolonged retention of the primary incisors (FDI 72, 82, and 81).

## 4. Discussion

The current study aimed to develop and evaluate a deep learning model for the automated segmentation and labelling of primary and permanent teeth in children's digital impressions obtained from IOS. ToothInstanceNet has been applied to combine large-context and high-resolution predictions for highly effective tooth instance segmentation and labelling. The results demonstrated nearly perfect tooth detection, and our proposed method outperformed all submissions to the 3DTeethSeg22 challenge by a significant margin. Therefore, these outcomes can serve as a benchmark to assess future algorithms for segmenting and labelling teeth in digital impressions of mixed dentitions.

Our findings demonstrate that segmenting digital impressions is a rapid and effective process. In dentistry, AI is expected to play a crucial role in accurately identifying dental elements, thereby increasing efficiency and reducing human error. Identifying primary and permanent teeth during the mixed dentition stage in digital impressions is especially challenging [17,19], mainly due to anatomical similarity between certain teeth (particularly canines), the limited colour differentiation in digital scans, and the variability in eruption timing and patterns, including atypical eruption sequences. Our deep learning solution overcame these difficulties, achieving excellent accuracy in distinguishing between different types of teeth and in accurately numbering them.

Accurate identification of teeth in digital impressions of children with mixed dentition is essential for several reasons. First, it serves as a foundational step for future AI-driven developments in more advanced applications involving IOS, such as tools for diagnosing caries, malocclusions, and dental anomalies. Integrating such a tool into IOS systems also enhances workflow efficiency, as automated identification of primary and permanent teeth in other imaging modalities has already demonstrated significant time savings in routine dental practice [18]. These tools additionally support improved visualisation, aiding in patient and caregiver education by making features of the digital impressions easier to understand and communicate [18,19]. However, it remains critical to emphasise the importance of data review by experienced professionals [29].

Therefore, following efficient segmentation, the next logical steps would involve AI-assisted diagnosis of dental conditions [30] and the automated design of treatment plans [31]. |For example, the developed algorithm may be used as a first step to objectively determine components of malocclusion for early orthodontic screening [32], resulting in more consistent early orthodontic referrals and the prevention of complex and costly treatments at a later stage [33]. Furthermore, the results of our method may be used to develop an algorithm for caries detection in primary teeth [3], identifying a major risk factor for caries in permanent teeth before they have even erupted [34]. A third application could investigate the automated detection of low-quality enamel, such as molar incisor hypomineralisation (MIH) [35]. This condition typically occurs during the mixed dentition stage, following the eruption of the permanent first molars and incisors, and early diagnosis can improve the management of post-eruptive enamel breakdown [36].

The proposed method outperformed other submissions in the 3DTeethSeg22 challenge [22], establishing a new benchmark for tooth

**Table 4**
Results of our method (ToothInstanceNet) and comparative methods based on the 3DTeethSeg22 challenge metrics. Most effective metrics are highlighted bold.

| Dataset | Method | Exp(-TLA) | TSA | TIR | Score |
|---|---|---|---|---|---|
| 3DTeethSeg22 | Chompers | 0.6242 | 0.8886 | 0.8795 | 0.7974 |
| | OS | 0.7845 | 0.9693 | 0.8940 | 0.8826 |
| | TeethSeg | 0.9184 | 0.9678 | 0.8538 | 0.9133 |
| | IGIP | 0.9244 | 0.9750 | 0.9289 | 0.9427 |
| | FiboSeg | **0.9924** | 0.9293 | 0.9223 | 0.9480 |
| | CGIP | 0.9658 | 0.9859 | 0.9100 | 0.9539 |
| | ToothInstanceNet | 0.9878 | **0.9862** | **0.9552** | **0.9764** |

Please see Section 2.5 or Ben-Hamadou et al. (2022) [22] for a description of each comparative method.

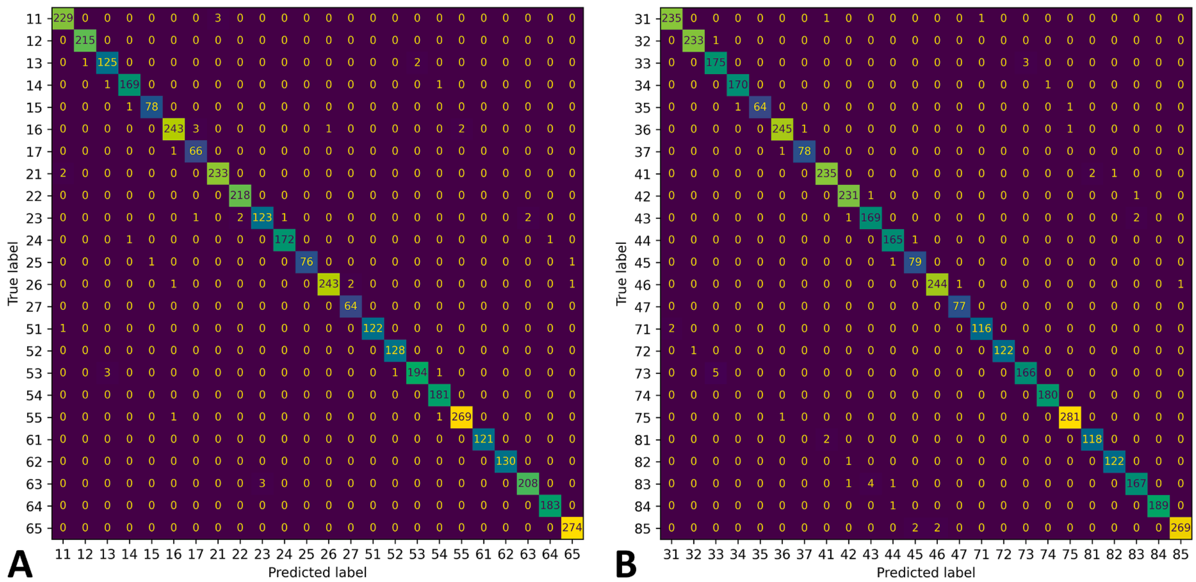


Fig. 2. Confusion matrices of tooth labelling for maxillary (A) and mandibular (B) teeth following the FDI tooth numbering system. The final method trained on the digital impressions with a primary or mixed dentition is evaluated. Each cell is coloured according to its value.

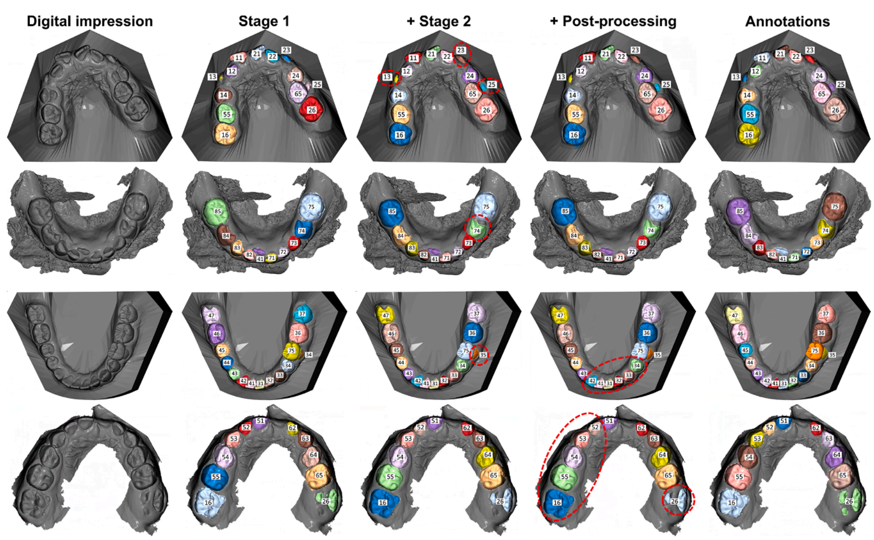


Fig. 3. Qualitative results for four representative digital impressions. The first column displays the input scan, while columns two to four show the predictions of our deep learning model after incorporating stage 1, stage 2, and post-processing, and the last column presents the reference annotations. Teeth are labelled using FDI numbers, and teeth inside a red dashed ellipse show more effective results compared to a previous model variant or compared to the annotations.

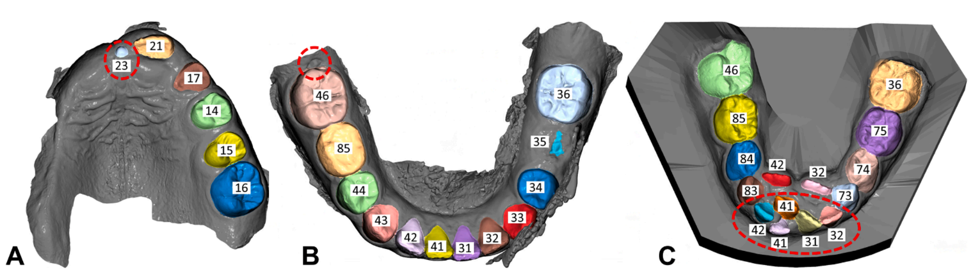


Fig. 4. Three failure cases showing (A) a false-positive detection, (B) a false-negative detection, and (C) incorrect FDI numbers in red dashed ellipses. Teeth are labelled using FDI numbers.

instance segmentation and labelling. Notably, the significant improvement in tooth labelling demonstrates the method's ability to integrate a large context for accurate FDI numbering. The challenge evaluation compared point-wise predictions with ground-truth annotations, favouring approaches like ours that directly predict point labels. However, the challenge did not include mesh-based methods that predict labels for mesh faces or triangles [37–40]. These approaches represent alternative data representations commonly used in dental segmentation tasks. As their evaluation frameworks and output formats differ from those adopted in our development, direct comparisons were not made.

Several point-based methods for tooth instance segmentation have been proposed. TSegNet implemented a two-stage method for tooth centroid prediction and single-tooth segmentation and labelling, respectively. The authors have trained and evaluated 2000 digital impressions, and reached good accuracy for these tasks [14]. Other authors proposed a method using 3D bounding boxes, and also obtained good results [41]. A third method called SRF-Net predicts FDI numbers of already segmented teeth and includes a label correction method to effectively process dental anomalies, achieving a macro-averaged accuracy higher than 0.9 for tooth labelling [42].

Other previous studies have revealed an increasingly higher accuracy for larger samples of digital impressions. In comparison to the current study, some published works included more digital impressions for training and evaluation [43,44]. However, the proposed algorithms were not developed for detecting and labelling primary teeth. Therefore, although the previous models are highly effective for permanent teeth, they are likely to experience reduced effectiveness for primary teeth due to a domain shift in the subject population [45]. As such, to integrate an automated method for tooth segmentation and labelling into daily dental practice, digital impressions of children must be considered during validation to ensure the algorithm applies to a wide range of subjects.

Automated segmentation and labelling of mixed dentitions have also been investigated in other image modalities, such as occlusal intraoral photographs [46], cone-beam CT scans [18,47], and panoramic radiographs [16,17,19,48]. These studies demonstrate that deep learning models can effectively segment and label mixed dentitions, regardless of the dental image modality under investigation. However, to our knowledge, this is the first AI development explicitly proposed for the automated segmentation and labelling of primary and permanent teeth in digital impressions. This is the main strength of our study. The current method can accurately differentiate between primary and permanent teeth and is highly effective for mixed dentitions, offering greater functionality than existing methods. In contrast, a previous study introduced a fully automated landmark recognition system for primary and permanent teeth, including tooth instance segmentation [49]. However, this system failed to detect 11 % of teeth and was limited to supporting either primary or permanent teeth exclusively.

To increase the number of digital impressions with mixed dentitions, this study incorporated scans from the train split of the 3DTeethSeg22 challenge dataset [21]. Additional corrections were made to enhance the dataset's quality during the annotation process. These included addressing missing teeth, refining imperfect tooth-gingiva borders, and rectifying incorrect FDI numbers for permanent teeth. These improvements likely contributed to the increased effectiveness of the method when applied to the unaltered test split. The findings of this study were directly compared to those of other challenge participants. Since the method was trained strictly on scans from the train split, any technique used to improve its effectiveness on the unaltered test split was considered valid.

Failure cases, highlight the model's limitations in interpreting digital impressions with unusual dental conditions. With a relatively small dataset size ($n = 716$), such cases were underrepresented, leaving the model unable to generalise effectively. Expanding the dataset to include more diverse cases, such as those involving cleft lip and palate [50], could help train a more robust and consistent model. This study has some limitations. Firstly, the currently developed method was not evaluated on digital impressions from an external centre, as each included dataset acquired scans from a unique age group. The generalizability of the results to digital impressions from an external centre should be tested in further studies. Secondly, recently proposed deep learning models for tooth instance segmentation and labelling were not compared to ToothInstanceNet for mixed digital impressions [14, 41–44]. Thirdly, the digital impressions from Nijmegen were acquired *in vitro* by scanning dental casts of primary dentitions. These scans may not represent digital impressions acquired in a clinical setting, as their quality is no longer affected by saliva or limited spacing [51]. Nevertheless, our model is expected to retain the currently reported results for digital impressions of primary dentitions acquired *in vivo*, as a large portion of our dataset was acquired in a clinical setting.

## 5. Conclusions

This study developed a highly effective deep-learning method for segmenting and labelling mixed dentitions in digital impressions. The proposed method almost perfectly detected all teeth and could process a digital impression in under two seconds on average. This fast and reliable segmentation technique, which differentiates between primary and permanent teeth, can significantly benefit daily dental practice by aiding in the documentation of tooth eruption in children during the mixed dentition stage.


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**Data availability statement:** The source code for this project will be made publicly available at https://github.com/nnistelrooij/3dteethland , including the revised annotations of the train split of the 3DTeethSeg22 challenge dataset, following publication.

## CRediT authorship contribution statement

**Niels van Nistelrooij:** Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis. **Haline Cunha de Medeiros Maia:** Writing – original draft, Validation, Resources, Data curation. **Lingyun Cao:** Writing – review & editing, Data curation. **Shankeeth Vinayahalingam:** Writing – review & editing, Supervision, Funding acquisition. **Bas Loomans:** Writing – review & editing, Resources. **Maximiliano Sergio Cenci:** Writing – review & editing, Validation. **Fausto Medeiros Mendes:** Writing – review & editing, Supervision, Project administration, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jdent.2025.105976.

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